

## **Introduction to Electronic Systems (4FTC2028)**

### **Regulated Variable Power Supply**

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# Introduction :

A regulated power supply (RPS) is a fundamental component in almost all electronic systems, ensuring that sensitive circuits receive a stable and reliable DC voltage regardless of fluctuations in the input power source. In this project, a complete RPS is designed, simulated, and implemented on a PCB Perfboard to convert a standard 220 V AC mains input into a regulated 5 V or 12 V DC output. These voltage levels are widely used in digital logic systems, microcontrollers, sensors, and various low-power electronic devices.

The aim of the project is to explore the full development process of a power supply, beginning with the theoretical design, followed by circuit simulation, and concluding with a practical hardware implementation. Key stages include AC step-down transformation, rectification, filtering, and voltage regulation. By constructing this system, the project enhances understanding of power electronics principles and provides hands-on experience with essential components such as transformers, diodes, capacitors, and voltage regulators.

# Components Used :

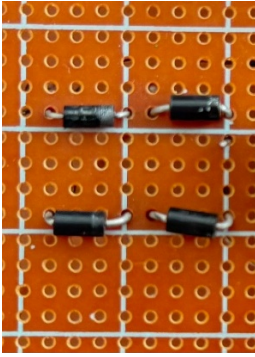
## 1. Transformer

The transformer reduces the 220V AC from the mains to a lower AC voltage that the circuit can safely use. It works by using two coils: the mains voltage goes into the primary coil, and a smaller voltage comes out of the secondary coil. The amount of voltage reduced depends on the number of turns in each coil. In this project, the transformer steps the voltage down to a low AC level that can then be rectified, filtered and regulated.



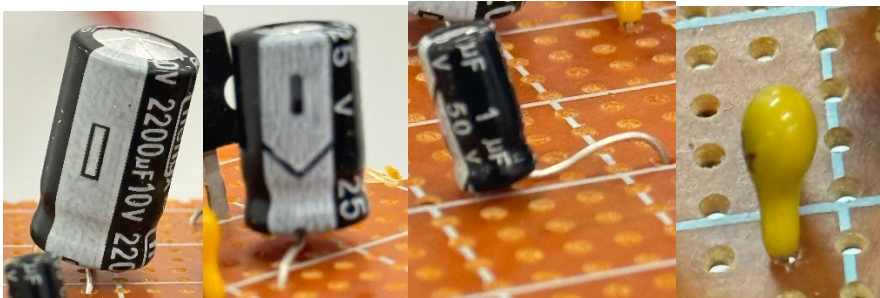
## 2. Full bridge rectifier

A full bridge rectifier is used in the power supply to convert the low voltage AC signal from the transformer into pulsating DC voltage. It consists of four diodes arranged in a bridge configuration. During each half-cycle of the AC input, two of the diodes conduct and allow current to pass in the same direction through the load. This means the rectifier uses both halves of the AC waveform, giving a smoother and higher DC output compared to a half-wave rectifier. This DC signal is then ready to be filtered and regulated in the next stages of the circuit.



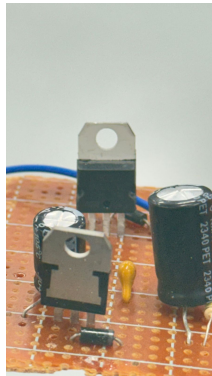
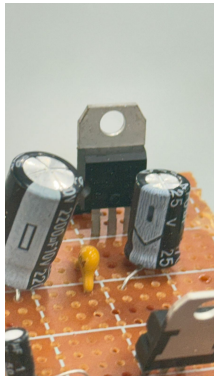
### 3. Capacitors

4 capacitors were used in our design to smooth and stabilise the DC outputs. The 2200  $\mu\text{F}$  and 470  $\mu\text{F}$  capacitors act as filter capacitors; they store charge and reduce the ripple in the rectified DC voltage, helping to produce a smoother output. The smaller 0.1  $\mu\text{F}$  capacitors are used for noise reduction and to improve the stability of the voltage regulator by filtering out high-frequency fluctuations. Together, these capacitors ensure the power supply delivers a clean and steady DC voltage.



### 4. Voltage Regulators

Voltage regulators are used to provide a stable and precise DC output regardless of changes in the input voltage or load. In this project, two types of regulators are used. The LM7805 is a fixed 5 V regulator that outputs a constant 5 V DC, suitable for low-voltage digital circuits. The LM317 is an adjustable regulator that allows the output voltage to be set to a desired level, such as 12 V, by using external resistors. These regulators ensure that the power supply delivers reliable and consistent DC voltages for different applications.



#### 5. Resistors

Resistors are used in the power supply to control and limit the flow of current in different parts of the circuit. In this project, they are mainly used with the LM317 regulator to set the output voltage by forming a voltage divider, and in other places to ensure components operate within safe current levels. They help keep the circuit stable and prevent excessive current from damaging components.

#### 6. Potentiometer

#### 7. Voltage sensor

